ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
GUWAHATI

Course Structure and Syllabus

(From Academic Session 2018-19 onwards)

B.TECH

ELECTRICAL ENGINEERING

7th Semester

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure
(From Academic Session 2018-19 onwards)

B.Tech 7th Semester: Electrical Engineering
Semester VII/ B.TECH/EE

Sl. No	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	EE181701	Power System-IV	3	1	0	4	30	70
2	EE1817PE3*	Program Elective-3	3	0	0	3	30	70
3	EE1817PE4*	Program Elective-4	3	0	0	3	30	70
4	EE1817OE3*	Open Elective-3	3	0	0	3	30	70
5	HS181704	Principles of Management	3	0	0	3	30	70
Practical								
1	EE181722	Project-1	0	0	6	3	50	50
2	SI181721	Internship-III (SAI-Industry)	0	0	0	2	-	200
TOTAL			**15**	**1**	**6**	**21**	**200**	**600**
Total Contact Hours per week : 22								
Total Credit: 21								

Program Elective-3		
Sl. No.	**Subject Code**	**Subject**
1	EE1817PE31	High Voltage Engineering
2	EE1817PE32	Computer Networks
3	EE1817PE33	Pattern Recognition and Machine Learning
4	EE1817PE34	Speech Coding Techniques
5	EE1817PE3*	Any other subject offered from time to time with the approval of the University

Program Elective-4		
Sl. No.	**Subject Code**	**Subject**
1	EE1817PE41	FACTS
2	EE1817PE42	Renewable Energy Sources
3	EE1817PE43	Advanced Electrical Drives
4	EE1817PE4*	Any other subject offered from time to time with the approval of the University

Open Elective-3		
Sl. No.	**Subject Code**	**Subject**
1	EE1817OE31	Optoelectronics
2	EE1817OE32	Computer Vision
3	EE1817OE33	Electromagnetic Waves
4	EE1817OE34	Electrical and Hybrid Vehicles
5	EE1817OE3*	Any other subject offered from time to time with the approval of the University

Detail Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
EE181701	**Power System-IV**	**3-1-0**	**4**

Prerequisites: Electric Power System –I, Electric Power System – II, Electric Power System - III

Course Objectives:

The objective of this course is to provide students with a broad understanding of electricity generation by conversion of various forms of energy to electrical energy and associated technology, operation and decision making on power plants. Provide a detailed study on the analysis and economic related issues in power sectors. The impact of this course offers a wide understanding on the different types of power plant, its functions and their flow lines and issues related to them

Course Outcomes:

CO1: Identify various characteristics of thermal and hydal plants and to solve various kinds of economic dispatch problems as well as UC problem

CO2: Make use of various modes of hydro-thermal scheduling and to prepare short term hydro thermal schedules

CO3: Comprehend the concept of Power System interconnection and economics of interconnected system

CO4: Apply generation control techniques in single area and multi area experimenting LFC, AVR and AGC

CO5: Utilize knowledge of State Estimation in power system and IEEE study model

MODULE 1: Power Plants- ***Thermal power plant*** – Economical, and ecological issues. Line diagram of thermal power plant Boiler: classifications and operational issues, Super heater, Reheaters, Economizer, air heater, draft system, feed water heater, and evaporator, cooling water supply and cooling tower. Speed governor, station auxiliaries, Ash handling system for thermal power plant. Emission issue of thermal power plant.

Hydro power plant- Classification of hydro power plant, economic and ecological issues related to the selection of site of hydro power plant, Estimation of generation capacity of hydro power plant, Selection of turbine, plant layout, governor and hydro plant auxiliaries.

Nuclear power plant- Fundamental of nuclear fission, schematic of nuclear power plant, nuclear fuels and fertile materials, nuclear reactor, chain reaction, moderator, coolants, control of fission, reactor operation, types of reactors, economical, and ecological issues of nuclear power plant.

Other power plants- Solar PV power plant, principle and operation, standalone Solar PV power plant, Grid Connected Solar PV power plant, Wind power plant etc.

MODULE 2: Economic Operation of Thermal plants- Methods of loading turbo-generators, input-output curves, heat ratio and incremental cost, coordination equation, economic loading of units, with and without transmission loss, penalty factor, iterative methods of solving co-ordination equation, economic thermal dispatching with network losses considered, B-matrix loss formula and its derivatives, economic dispatch versus unit commitment(UC), constraints in UC, UC solution method, generation Scheduling, ,introduction to load forecasting.

MODULE 3: Automatic load frequency and voltage control- Concept of load frequency control (LFC), LFC model for single/multi area systems, tie-line control in interconnected system (multi area system), concept of voltage & reactive power control.

Textbooks/ Reference Books:

1. P.K. Nag, Power Plant Engineering, 3/e, TMH.
2. GK Nagpal, Power Plant Engineering, Khanna Publishers.
3. B.R. Gupta, Generation of Electrical Energy, S Chand and Company.
4. M.V. Despande, Elements of Electrical Power System Design, A.H. Wheeler.
5. S.L. Uppal, Electrical Power, Khanna Publishers.
6. Prabha KundurPower System stability & control

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817PE31	**High Voltage Engineering**	**3-0-0**	**3**

Course Objective: To introduce theory of high voltage breakdown phenomena and testing of insulation strength of dielectrics

Course outcomes: At the end of the course the students will be able to:

- **CO1:** Analyze breakdown of dielectrics
- **CO2:** Design generation of high voltage for testing insulation strength
- **CO3:** Analyze performance of high voltage testing and measurement methods

MODULE 1: Breakdown of Gasses

Basic processes in gas breakdown, generation of electron avalanche, Townsend mechanism, secondary processes, criterion for spark breakdown, Streamer mechanism, time lag for spark breakdown, breakdown in electro negative gasses.

MODULE 2: Breakdown in Liquid Dielectrics

Liquid dielectric, origin, electrical conduction and breakdown phenomena of pure liquids, test cells, natural and induced conduction, Transformer oil; properties, purification.

MODULE 3: Breakdown in Solid Dielectrics

Intrinsic and related forms of breakdown in solids, measurement of intrinsic strength- preparation of the specimen, surface discharge phenomenon, positive and negative Lichtenbury patterns, discharge detection, breakdown of solid insulation by Tracking, lightening phenomena-nature and characteristic of lightening induced voltage, measuring instruments and devices, lightening protection.

MODULE 4: **Generation of High Voltage**

High voltage testing transformers, cascaded transformers, high frequency transformers, tesla coil, cascaded rectifier for DC high voltage, electrostatic generators, impulse generator; analysis of the basic circuit, multi-stage circuits, factors influencing design, general construction, triggering, synchronization of operation, delay cables.

MODULE 5: High Voltage Testing

Low frequency direct current, high frequency and impulse tests, terminology; disruptive discharge, flashover and puncture, withstand voltage, 50% disruptive discharge voltage, impulse wave, full and chopped wave, impulse ratio, volt-time curve; testing of overhead insulators, bushings, cables, arrangement of the test object, various tests and test conditions.

MODULE 6: High Voltage Measurements and Testing: Electrostatic voltmeters, potential dividers; resistive, capacitive and mixed resistive dividers, peak voltage measuring device, oscillographic measurements, Schering bridge, sphere and rod gaps. Types of High Voltage Tests; High Voltage Testing (Impulse) Transformers; Voltage Control by Variation of Alternator Field Current, Tapped Transformers; Induction Regulators; Control Gear and Protective Devices; Equipment for Voltage Measurement, Measurement of R.M.S., Peak, and Instantaneous Values of Voltages; Low Frequency High Voltage Tests; High Voltage D.C. Tests; High Voltage D.C. Testing of Cables; Localization of Faults in H.V. Cables; High Frequency H.V. Testing; Surge Testing; Basic Impulse Generator; Testing of Insulating Material; Impulse Testing of Transformer; H.V. Testing of Cables; Testing of Strength of Insulating Oils; High Voltage Testing of Porcelain Insulators.

MODULE 7: Insulation Design Principle

Classification of insulating materials, material of dielectric in parallel, dielectric hysteresis, high voltage cables, thermal and electrical cables, requirements of cable design, materials used, high voltage bushing types, basic applications, insulation of high voltage transformers.

MODULE 8: High Voltage Laboratory

HV plants and apparatus, HV connections, laboratory earthing and safety measures.

Textbooks/ Reference Books:

1. High Voltage Technology 2006 - L. L. Alston, Oxford University Press
2. Electrical Breakdown of Gases 1978 - J.M. Meek and J.D. Craggs, John Wiley & Sons
3. High Voltage Engineering 5th Ed 2013 – M. S. Naidu and V Kamaraju, McGraw Hill
4. High Voltage Engineering 3rd Ed 2012 – C. L. Wadhwa, New Age International
5. High Voltage Engineering Fundamentals 2nd Ed 2008 – E. Kuffel, W. S. Zaengl and J. Kuffel, Elsevier
6. A Course in High Voltage Engg- R. S. Jha, Dhanpat Rai
7. An Introduction to High Voltage Experimental Technique 1978 – D. Kind, John Wiley & Sons

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817PE32	**Computer Networks**	**3-0-0**	**3**

COURSE OBJECTIVES:

- To teach the basics of the working of the Internet.
- To teach fundamentals of the protocols involved in the working of various layers of the Internet.

COURSE OUTCOMES:

At the end of this course, the students will be able:

CO1: To Understand the Basics of the Working of the Internet.

CO2: To Study Various Internet Protocols.

CO3: To Understand Various Security Measures in Internetworking.

CO4: To Understand Various Uses of Computer Networking.

MODULE 1: Introduction to Internet

a) To study Topology of Internet.
b) To study structure of Internet, Internet standards and Internet administration.
c) To study protocol layering.

MODULE 2: The Physical Layer

a) To learn Information Theory.
b) To learn coding of data, conversion methods between various types of data and multiplexing techniques.
c) To learn about various transmission media.
d) To learn about Circuit Switching, Packet Switching and various switches.

MODULE 3: The Data Link Layer

a) To learn error detection and correction techniques.
b) To learn DLC protocols.
c) To learn MAC protocols.
d) To learn various wired and wireless LAN protocols.
e) To learn mobile telephone protocols.
f) To learn satellite systems.

MODULE 4: **The Network Layer**

a) To learn packet switching methods.
b) To learn congestion control techniques.
c) To learn IP, IPv4, ICMPv4,
d) To learn Unicast Routing and Multicast Routing
e) To learn about IPv6 and ICMPv6,

MODULE 5: The Transport Layer

a) To learn about transmission layer issues protocols.
b) To learn about UDP, TCP-IP, SCTP

MODULE 6: The Application Layer

a) To learn basics of WWW, HTTP, FTP, E-Mail and DNS
b) To learn issues concerning multimedia in Internet.
c) To learn basics of Internet Security.

Textbooks/ Reference Books:

1. Computer Networking:: Stallings :: TMH
2. Computer Networks:: Tannenbaum :: PHI
3. Data Communication and Networking :: Forouzan :: McGraw Hill India

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817PE33	**Pattern Recognition and Machine Learning**	**3-0-0**	**3**

COURSE OBJECTIVES:
To teach concepts of inculcating artificial intelligence in machines by different computational methods

COURSE OUTCOMES: At the end of this course, the students will be able to
CO1: describe structure, components and mathematical formulation of learning by machines
CO2: develop algorithms for learning and pattern recognition by machines
CO3: analyze performances and characteristics of machine learning and pattern recognition algorithms for utilization in societal, academic and industrial purposes

MODULE 1: Introduction to Machine Learning
Intelligent Machines, Data Representation, Diversity of Data, Forms of learning, Machine learning and data mining, Linear Algebra and Machine Learning Techniques

MODULE 2: Supervised Learning
Learning from observation, Bias and Variance, Computational Learning Theory, Estimating Generalization errors, Metrics for assessing regression, Metrics for assessing classification

MODULE 3: Statistical Learning
Descriptive Statistics in learning Techniques, Bayesian Reasoning, k-Nearest Neighbour (kNN) Classifier, Discriminant functions and Regression functions, Linear Regression with Least Square Error criterion, Logistic Regression for Classification Tasks

MODULE 4: Learning with Support Vector Machine (SVM)
Linear Discriminant Function for binary classification, Linear Maximal Margin classifier for overlapping classes, Non Linear classifier, Regression by SVM, Variants of basic SVM Techniques

MODULE 5: Learning with Neural Networks
Cognitive Machines, Neuron Models, Network Architectures, Perceptrons, Linear Neurons, Error correction delta Rule, Multilayer perceptron (MLP) networks, Radial Basis Function (RBF) networks, Genetic Neural Systems

MODULE 6: Fuzzy Inference System
Cognitive Uncertainty and Fuzzy Rule Base, Fuzzy Quantification of Knowledge, Fuzzy Rule Base and Approximate Reasoning, Mamdani Model, Neuro Fuzzy Inference model, Genetic Fuzzy Systems

MODULE 7: Data Clustering and Data Transformation
Unsupervised Learning, Engineering Data, Overview of basic clustering models, K-Means Clustering, Fuzzy k-means clustering, some useful data transformation, Introduction to PCA, Decision Tree Learning, Measuring of Impurity for evaluating Splits in Decision Trees, Pruning Decision Trees, Strengths and weaknesses of Decision Tree Approach

MODULE 8: Application of Machine Learning
Some Practical Application Implementation

Textbooks/ Reference Books:

1. M Gopal: Applied Machine Learning, McGraw Hill Education
2. Geoffry J. McLachlan: Discriminant Analysis and Statistical Pattern Recognition, John Wiley & Sons
3. Richard O. Duda, Peter E. Hart, David G. Stork: Pattern Classification, John Wiley & Sons
4. S. Theodoridis, K. Koutroumbas: Pattern Recognition, Elsevier
5. Keinosuke Fukunaga: Introduction to statistical pattern recognition, Morgan Kaufmann, Academic Press
6. Shai Shalev-Shwartz, Shai Ben-David: Understanding Machine Learning: From Theory to Algorithms Cambridge University Press, 2015
7. S. Rajasekaran, G. A. Vijayalakshmi Pai: Neural Networks, Fuzzy Systems and Evolutionary Algorithms: Synthesis and Applications, PHI 2017.
8. B. Yegnanarayana: Artificial Neural Networks, PHI 2015
9. Simon Haykin: Neural Networks, Pearson Education, 2003.
10. Laurance Fausett: Fundamentals of Neural Networks, Englewood cliffs, N.J., Pearson Education, 1992.
11. Jacek M. Zurada: Introduction to Artificial Neural Systems, Jaico Publishing Home, 2002.
12. U Dinesh Kumar Manaranjan Pradhan: Machine Learning using Python, John Wiley & Sons 2019
13. B. Kosko: Neural Networks and Fuzzy Systems, Prentice-Hall of India Pvt. Ltd., 1994.
14. G. J. Klir and T. A. Folger: Fuzzy Sets, Uncertainty and Information, Prentice-Hall of India Pvt. Ltd., 1993.
15. H.J. Zimmermann: Fuzzy Set Theory and Applications, Allied Publication Ltd., 1996.
16. Timothy J. Ross: Fuzzy Logic with Engineering Applications, Tata McGraw Hill, 1997
17. John Yen & Reza Langari: Fuzzy Logic – Intelligence Control & Information, Pearson Education, New Delhi, 2003.
18. Driankov, Hellendroon: Introduction to Fuzzy Control, Narosa Publishers.
19. David Goldberg: Genetic Algorithms and Machine learning, PHI
20. Z Michalewicz: Genetic Algorithms + Data Structures = Evolution Programs, 3rdEd, Springer, 1996.
21. T Baeck, D B Fogel, Z Michalewicz: Evolutionary Computation Vol 2 Advanced Algorithms and Operators, Institute of Physics Publishing, Bristol, UK, 2000
22. Y. Gong, W. Xu: Machine Learning for Multimedia Content Analysis, Springer
23. Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar: Foundations of Machine Learning, MIT Press.
24. Simon Haykin: Neural Networks & Learning Machines, Pearson 2016

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817PE34	**Speech Coding Techniques**	**3-0-0**	**3**

Pre-requisite: Digital Signal Processing

Course Outcomes: At the end of the course the student will be able to

1. Model the speech signal
2. Analyze the quality and properties of speech signal.
3. Analyze quality of the processed speech signals.

MODULE 1: Speech Production and Modeling

Human Auditory System; General structure of speech coders; Classification of speech coding techniques – parametric, waveform and hybrid; Requirements of speech codecs–quality, coding delays, robustness

MODULE 2: Speech Signal Processing

Pitch-period estimation, all-pole and all-zero filters, convolution; Power spectral density, periodogram, autoregressive model, autocorrelation estimation

MODULE 3: Linear Prediction of Speech

Basic concepts of linear prediction; Linear Prediction Analysis of non-stationary signals –prediction gain, examples; Levinson-Durbin algorithm; Long term and short-term linear prediction models; Moving average prediction.

MODULE 4: Speech Quantization

Uniform quantizer, optimum quantizer, logarithmic quantizer, adaptive quantizer, differential quantizers, Adaptive Huffman coding, Arithmetic coding, LZW coding, Psychoacoustic model, Perceptual coding and masking techniques;

Vector quantization – distortion measures, codebook design, codebook types

MODULE 5: Scalar Quantization of LPC

Spectral distortion measures, Quantization based on reflection coefficient and log area ratio, bit allocation; Line spectral frequency – LPC to LSF conversions, quantization based on LSF

MODULE 6: LPC Model

LPC model of speech production; Structures of LPC encoders and decoders; Voicing detection; Limitations of the LPC model

MODULE 7: Code Excited Linear Prediction

CELP speech production model; Analysis-by-synthesis; Generic CELP encoders and decoders; Excitation codebook search – state-save method, zero-input zero-state method; CELP based on adaptive codebook, Adaptive Codebook search; Low Delay CELP and algebraic CELP

MODULE 8: Speech Coding Standards

An overview of ITU-T G.726, G.728, G.729, MPEG Audio layers, Dolby AC3 standards

Textbooks/ Reference Books:

1. Speech Coding Algorithms: Foundation and Evolution of Standardized Coders - W.C. Chu, Wiley Inter science, 2003.
2. Fundamentals of Acoustics 4th Edition – L. E. Kinsler, A. R. Frey, A. B. Coppens, J. V. Sanders, John Wiley & Sons, 2005.
3. Discrete-Time Speech Signal Processing - Thomas F. Quatieri, Pearson Education, 2002.
4. Fundamentals of Speech Recognition – L. Rabiner, B-H Juang, B. Yegnanarayana, Pearson.
5. Digital Processing of Speech Signals – L. R. Rabiner, R. W. Schafer, Pearson, 1993.
6. Speech Communication Human and Machine – D O 'Shaughnessy, Universities Press (India) Pvt Ltd, IEEE2001.
7. Speech and Language Processing - Daniel Jurafsky & James H.Martin, Pearson Education, 2000.
8. Spoken Language Processing - Xuedong Huang, Alex Acero, Hsiad, Wuen Hon, Prentice Hall, 2001.
9. Computer Speech – Recognition, Compression, Synthesis - M.R.Schroeder, Springer Series in Information Sciences, 1999.
10. Discrete Time Signal Processing - A. V. Oppehein and R.W. Schafer, PHI, 1992.
11. Digital Speech by A. M. Kondoz, 2nd Edition, Wiley Students Edition, 2004.
12. Speech and Audio Signal Processing - B.Gold and N.Morgan, John Wiley & Sons, 2000

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817PE41	**FACTS**	**3-0-0**	**3**

Pre-requisite:
Knowledge of power electronics based converters, various types of modulation techniques etc.

Course Outcomes
At the end of the course, the students will be able to:
CO1: Explain working principles of FACTs devices, their applications etc.
CO2: Model various FACTs devices and to implement them in various existing algorithms
CO3: Analyze the characteristics of AC transmission and the effect of shunt and series reactive compensation
CO4: Explain working and protection of various HVDC system
CO5: Design HVDC substation, three phase full wave bridge converters of various pulses

MODULE 1: FACTs Devices
Description and characteristics of thyristor based FACTs devices, their representation, opportunities of FACTs, Basic types of FACTS controllers, Relative importance of different types of controllers, Models of FACTs devices and their implementation in various existing algorithms. Basic concepts of voltage sourced converters (VSC), single phase full wave bridge converter operation, single phase pole operation.

MODULE 2: Phase Shifters
Principle and operation of single phase phase shifter, steady state model of a static phase shifter (SPS), power circuit configuration of SPS, SPS applications.

MODULE 3: Series Compensation and Shunt Compensation
Steady state voltage regulation and prevention of voltage collapse, power flow control, series compensation schemes, working principle, model and application of TCSC and SSSC, principle of shunt compensation, effect of shunt compensation of power angle diagram, principle of operation, configuration and control of SVC, STATCOM configuration and control, application of SVC, STATCOM etc.

MODULE 4: UPFC
UPFC: Its working principle, construction, control, uses etc.

MODULE 5: DC Transmission Technology
Comparison of AC and DC Transmission (Economics, Technical Performance and Reliability); Application of DC Transmission; Types of HVDC Systems; Components of a HVDC system. Line Commutated Converter and Voltage Source Converter based systems.

MODULE 6: Analysis of Line Commutated and Voltage Source Converters
Line Commutated Converters (LCCs): Six pulse converter; Analysis neglecting commutation overlap, harmonics, Twelve Pulse Converters; Inverter Operation; Effect of Commutation Overlap; Expressions for average dc voltage, AC current and reactive power absorbed by the converters; Effect of Commutation Failure, Misfire and Current Extinction in LCC links

Textbooks/ Reference Books:

1. N G Hingorani, L Gyugyi—Understanding FACTS, IEEE Press
2. Y H Song, Allan T Jones---Flexible AC Transmission Systems, IEE Book
3. K.R. Padiyar,–HVDC Power Transmission Systems, New Age International Publishers, 2011.
4. S. Rao, –EHV-AC, HVDC Transmission and Distribution Engineering‖, Khanna Publishers
5. J. Arrillaga, –High Voltage Direct Current Transmission‖, Peter Peregrinus Ltd., 1983.
6. E. W. Kimbark, –Direct Current Transmission‖, Vol.1, Wiley-Interscience, 1971.
7. S.L. Uppaland S. Rao, –Electrical Power Systems‖, Khanna Publishers

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817PE42	**Renewable Energy Sources**	**3-0-0**	**3**

Objectives

Introduce fundamental concepts in Renewable Energy Sources, advantages and disadvantages, design, simulation and their applications.

Course Outcomes

CO1: To comprehend the world energy situation and the notion of distributed end use energy and to understand the bad effects of the present concentration use of energy.

CO2: To understand the different types of renewable energy sources, their advantages/disadvantages and applications

CO3: To able to know the basics of solar energy and to be able to design and development of solar photovoltaic/thermal systems.

CO4: To be able to model, analyze and design wind energy systems along with biomass based systems.

CO5: To be able to understand and analyze the energy from the ocean wave, Magneto-Hydro-Dynamic Generation and fuel cell.

MODULE 1: Introduction

Fossil fuel based systems, impact of fossil fuel based systems, renewable energy – sources and features, seasonal variations and availability, importance, primary & secondary energy sources, limitations to primary sources, various sources of renewable energy, applications

MODULE 2: Solar Energy Solar Geometry

Solar radiation, solar radiation angles, local solar time, solar radiation spectrum, radiation measurement, solar collector-flat plate collector & solar concentrator, solar heater-water heater & air heater, solar cooker, solar distillation, solar energy storage- sensible heat storage & latent heat storage.

MODULE 3: Solar Photovoltaic Systems

Operating principle, photovoltaic cell concepts, cell, module, array, series and parallel connections, Maximum power point tracking (MPPT)

MODULE 4: Wind Energy

Wind turbine rotor -classification, characteristics, Analysis of ideal wind turbine rotor, Power co-efficient, Types of wind mills, Site selection Characteristics of wind generators

MODULE 5: Biomass

Operating principle, classification, design and applications

MODULE 6: Energy from the Ocean

Tidal energy, wave energy, ocean thermal energy conversion (OTEC) introduction, types, plants & their specifications

MODULE 7: Geo-Thermal Energy

Sources and use of geo-thermal energy, classification of geo-thermal power plants

MODULE 8: Magneto Hydro Dynamic

Generation Principles of MHD generation, MHD generator, equivalent circuits, MHD system

MODULE 9: Fuel Cell

Introduction, energy conversion principles, types of fuel cell, components of a fuel cell, polarization

Textbooks/ Reference Books:

1. Swami Saran, "Soil Dynamics and Machine Foundations", Galgotia Publications Pvt. Ltd., New Delhi.
2. Shamsher Prakesh and Vijay Kumar Puri, "Foundations for Machines: Analysis and Design", A Wiley-Interscience Publication, John Wiley and Sons.
3. P. Srinivasulu and C. V. Vaidyanathan, "Hand Book of Machine Foundations", McGraw-Hill Education.
4. Steven L. Kramer, "Geotechnical Earthquake Engineering", Prentice Hall International Series, Pearson Education India
5. F. E. Richart, Jr., J. R. Hall, Jr. and R. D. Woods, "Vibrations of Soils and Foundations", Prentice-Hall International Series
6. IS 2974-1: "Code of Practice for Design and Construction of Machine Foundations", Part 1: Foundation for Reciprocating Type Machines, Bureau of Indian Standards

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817PE43	**Advanced Electrical Drives**	**3-0-0**	**3**

Course Outcomes: At the end of this course the student will be able to:

1. Analyze the operation of power electronic converters and their control strategies
2. Analyze the vector control strategies for ac motor drives
3. Implement efficient control strategies for ac motor drives

MODULE 1: Power Converters for AC drives

PWM control of inverter, selected harmonic elimination, space vector modulation, current control of VSI, three level inverter, Different topologies, SVM for 3 level inverter, Diode rectifier with boost chopper, PWM converter as line side rectifier, current fed inverters with self- commutated devices. Control of CSI, H bridge as a 4-Q drive

MODULE 2: Induction Motor Drives

Different transformations and reference frame theory, modeling of induction machines, voltage fed inverter control-v/f control, vector control, direct torque and flux control (DTC)

MODULE 3: Synchronous Motor Drives

Modeling of synchronous machines, open loop v/f control, vector control, direct torque control, CSI fed synchronous motor drives

MODULE 4: Permanent Magnet Motor Drives

Introduction to various PM motors, BLDC and PMSM drive configuration, comparison, block diagrams, Speed and torque control in BLDC and PMSM

MODULE 5: Switched Reluctance Motor Drives

Evolution of switched reluctance motors, various topologies for SRM drives, comparison, Closed loop speed and torque control of SRM

MODULE 6: DSP Based Motion Control

Use of DSPs in motion control, various DSPs available, realization of some basic blocks in DSP for implementation of DSP based motion control

Textbooks/ Reference Books:

1. P. C. Krause, O. Wasynczuk and S. D. Sudhoff: Analysis of Electric Machinery and Drive Systems, John Wiley & Sons, 2013
2. H. A. Taliyat and S. G. Campbell: DSP based Electromechanical Motion Controlǁ, CRC press, 2003
3. R. Krishnan: Permanent Magnet Synchronous and Brushless DC motor Drivesǁ, CRC Press, 2009
4. B. K. Bose: Modern Power Electronics and AC Drivesǁ, Pearson Education, Asia, 2003

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817OE31	**Optoelectronics**	**3-0-0**	**3**

Course Objectives

- To know the basics of solid-state semiconductor physics and optical fiber wave-guides and understand the nature and characteristics of light.
- To study the performance parameters of optical source and detector and to understand different LED and laser types and their applications.
- To learn the principle of optical detection mechanism in different detection devices.
- To understand different light modulation techniques and the concepts and applications of
- optical switching.
- To become familiar with different optical sensors.

Course Outcomes

CO1: To review solid-state semiconductor physics and optical fiber wave-guides.
CO2: To understand the different types of optical sources, their advantages/disadvantages and applications.
CO3: To able to know the basics of different optical detection devices
CO4: To be able to distinguish among different light modulation techniques and electro optic effects
CO5: To be able to design and development of optical sensors

MODULE 1: Optical Fiber Wave Guides

Theory of Dielectric slab waveguides-Symmetric and Asymmetric slab wave guide, Channel waveguide, Review of ray theory-Electromagnetic mode theory-Phase and group velocity-Modes-guided, radiative and leaky modes-'V' number-cut off wave length-Step index and graded index fibers-Parameters of optical fiber-problems. Signal degradation in fibers-Attenuation-Absorption loss-Linear and nonlinear scattering loss-Fiber bend loss-Dispersion mechanisms-Intramodal and intermodal dispersion-Expressions-modal noise-overall dispersion in single mode/multimode fibers-problems-mode coupling.

MODULE 2: Optical Sources

Light emitting diodes- P N junction characteristics- Direct and Indirect band gap materials-Spontaneous emission- Carrier concentration variation in n+p junction- carrier life time-Diffusion coefficient- Diffusion length- Injection efficiency- internal Quantum efficiency-Power internally generated- Overall efficiency of LED- problems- Heterojunction LEDs – Advantages- LED modulation- Electrical and Optical Bandwidth- LED structures-ELEDs and SLEDs-LED characteristics-Effect of temperature- LED Drive Circuits.
LASER diodes- Spontaneous Vs Stimulated emission-Einstein's relation-population inversion-cavity resonance and threshold gain-Laser modes-stimulated emission in PN junction-Rate equation-condition for lasing-Laser diode characteristics-Modulation-frequency chirp Heterojunction LASER-LASER structures-LED Vs LASER diodes.

MODULE 3: Optical Detectors

Optical Detectors and Fiber optic link, classification of detectors-Photodiodes-PN junction photo diode-PIN photodiode- response and noise- APDs –Advantages of APD- APD Bandwidth and noise-Phototransistor-parameters of phototransistor-problems-Detector performance parameters-noises- NEP

Power launching and coupling- source to fiber coupling-joints- fiber to detector coupling- losses-fiber splicers, connectors and couplers-types-Fiber optic link-System considerations-link power budget-rise time budget-Link Design

MODULE 4: Electro Optic Effects
Birefringence phenomenon EO Retardation, EO Amplitude and Phase Modulator, Electro optic Intensity Modulators, Beam deflection, Acousto-optics, A-O Modulators, Integrated optic spectrum analyzer, Nonlinear optics second harmonic generation, Parametric amplification. Advanced system Technology-Optical Amplifiers-Raman and Erbium doped optical Amplifiers-Noises-Wave Length Division Multiplexing (WDM) and Components-Optical network-wave length routed networks.

MODULE 5: Optical Fiber Sensors
Fiber optic sensors-classification, Multimode fiber Sensors-Displacement, pressure, stress, strain. Intensity modulated sensors, Active multimode FO sensors, Micro-bend optical fiber sensor, Current sensors, Magnetic sensors, Single mode FO sensors, Phase modulated, Polarization modulated, Fiber Optic Gyroscope. Fiber bragg gratings for strain and temperature sensors-displacement sensor-optical computing concepts-optical logic gates.

Textbooks/ Reference Books:

1. B.H Khan, Non-Conventional Energy Sources, Tata Mc Graw Hill Pvt. Ltd.
2. G.D. Rai, Nonconventional energy sources, Khanna publishers
3. S.Hasan Saeed and D.K Sharma, Non-Conventional Energy Resources, Katson Books
4. S.K Dubey and S.K Bhargava, Non-Conventional Energy Resources, Dhanpat Rai & Co.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817OE32	**Computer Vision**	**3-0-0**	**3**

Prerequisite: Digital Signal Processing as Open Elective-1, Digital Image Processing as Open Elective-2

COURSE OBJECTIVES

- This course is an introduction to the fundamental concepts and techniques in computer vision and their applications to solve real life problems

COURSE OUTCOMES: After completion of the course students will be able to
CO1: analyze mathematical formulation of 3D model image model and its coordinate system
CO2: analyze the algorithms of determining shape and depth of object from its image(s)
CO3: analyze performance of methods of determining shape and depth of object from its image(s)

MODULE 1: Image Formation
Radiometric Properties of Light Sources; Qualitative Radiometry; Sources and their Effects; Point Sources; Line Sources; Area Sources; Gourad & Phong Shading models; Shadows; Ambient Illumination; Photometric Stereo; Normal and Albedo from Many Views; Shape from Normals; Interreflections: Global Shading Models.

MODULE 2: Image Models
Coordinate Systems and Homogeneous Coordinates; Perspective Projection

MODULE 3: Shape from Texture
Representing Texture; Extracting Image Structure with Filter Banks; Analysis (and Synthesis) Using Oriented Pyramids; The Laplacian Pyramid; Shape from Texture: Planes and Isotropy; Recovering the Orientation of a Plane from an Isotropic Texture; Recovering the Orientation of a Plane from a Homogeneity Assumption; Shape from Texture for Curved Surfaces; Shape from Texture

MODULE 4: Geometry of Multiple Views
Two Views; Epipolar Geometry; Three Views; Trifocal Geometry; More than 3 Views; Stereopsis: Reconstruction; Camera Calibration; Image Rectification; Human Vision: Stereopsis; Binocular Fusion; Correlation; Multi-Scale Edge Matching; Dynamic Programming; Using More Cameras; Trinocular Stereo; Multiple-Baseline

MODULE 5: Affine Structure from Motion
Elements of Affine Geometry; Affine Structure from Two Images; The Affine Structure-from-Motion Theorem; Regularization theory; Optical computation; Optical flow; Motion estimation

MODULE 6: Projective Structure from Motion
Motion Estimation from Trifocal Tensors; Motion Estimation from Multiple Views

Textbooks/ Reference Books:

1. Richard Szeliski: Computer Vision: Algorithms and Applications, Springer, 2011
2. David A. Forsyth, Jean Ponce: Computer Vision A Modern Approach, Pearson 2015

3. Manas Kamal Bhuyan: Computer Vision and Image Processing: Fundamentals and Applications, CRC Press, 1st Edition, 2019
4. Salman Khan, Hossein Rahmani, Syed Afaq Ali Shah, Mohammed Bennamoun: A Guide to Convolutional Neural Networks for Computer Vision, Morgan & Claypool Publishers 2018
5. Milan Sonka, Vaclav Hlavac, Roger Boyle: Image Processing Analysis and Machine Vision and MindTap, Cengage India Private Limited; Fourth edition 2017
6. Adrian Kaehler, Gary Bradski: Learning OpenCV 3 - Computer Vision In C++ With The OpenCV Library, Shroff/O'Reilly 2017
7. Miguel Aranda, Gonzalo López-Nicolás, Carlos Sagüés: Control of Multiple Robots Using Vision Sensors, Springer 2017
8. Jan Erik Solem: Programming Computer Vision with Python: Tools and algorithms for analysing images, Shroff/O'Reilly 2012
9. Bernd Jahne, H. Haubecker: Computer Vision and Applications A Guide for Students and Practitioners, Elsevier 2006
10. Richard Hartley, Andrew Zisserman: Multiple View Geometry in Computer Vision, Cambridge University Press 2004
11. Stephan Heuel: Uncertain Projective Geometry: Statistical Reasoning for Polyhedral Object Reconstruction, Springer 2004
12. Berthold Klaus Paul Horn: Robot Vision, MIT Press, McGraw-Hill 1986
13. Dana H. Ballard and Christopher M. Brown: Computer Vision 2ndEd PHI 1982

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817OE33	Electromagnetic Waves	3-0-0	3

COURSE OBJECTIVES

- This course is an introduction to the fundamental concepts of radiation, transmission or propagation and reception of electromagnetic waves

COURSE OUTCOMES: After completion of the course students will be able to

CO1: describe formulation mathematical models of radiation, transmission or propagation and reception of electromagnetic waves

CO2: analyze transmission or propagation of electromagnetic waves through conductor and dielectric media

CO3: analyze radiation from antenna and transmission or propagation of electromagnetic waves through wave guides

MODULE 1: Transmission Lines

Concept of distributed elements, Equations of voltage and current, Standing waves and impedance transformation, Lossless and low-loss transmission lines, Power transfer on a transmission line, Analysis of transmission line in terms of admittances, Transmission line calculations with the help of Smith chart, Applications of transmission line, Impedance matching using transmission lines.

MODULE 2: Maxwell's Equations

Basic quantities of Electromagnetics, Basic laws of Electromagnetics: Gauss's law, Ampere's Circuital law, Faraday's law of Electromagnetic induction. Maxwell's equations, Surface charge and surface current, Boundary conditions at media interface.

MODULE 3: Uniform Plane Wave

Homogeneous unbound medium, Wave equation for time harmonic fields, Solution of the wave equation, Uniform plane wave, Wave polarization, Wave propagation in conducting medium, Phase velocity of a wave, Power flow and Poynting vector.

MODULE 4: Plane Waves at Media Interface

Plane wave in arbitrary direction, Plane wave at dielectric interface, Reflection and refraction of waves at dielectric interface, Total internal reflection, Wave polarization at media interface, Brewster angle, Fields and power flow at media interface, Lossy media interface, Reflection from conducting boundary

MODULE 5: Waveguides

Parallel plane waveguide: Transverse Electric (TE) mode, transverse Magnetic (TM) mode, Cut-off frequency, Phase velocity and dispersion; Transverse Electromagnetic (TEM) mode, Analysis of waveguide-general approach, Rectangular waveguides

MODULE 6 Antennas

Radiation parameters of antenna, Potential functions, Solution for potential functions, Radiations from Hertz dipole, Near field, Far field, Total power radiated by a dipole, Radiation resistance and radiation pattern of Hertz dipole, Hertz dipole in receiving mode.

Textbooks/ Reference Books:

1. R. K. Shevgaonkar: Electromagnetic Waves, Tata McGraw Hill, 2005
2. D. K. Cheng: Field and Wave Electromagnetics, Addison-Wesley, 1989
3. M. N.O. Sadiku: Elements of Electromagnetics, Oxford University Press, 2007
4. C. A. Balanis: Advanced Engineering Electromagnetics, John Wiley & Sons, 2012
5. C. A. Balanis: Antenna Theory: Analysis and Design, John Wiley & Sons, 2005

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1817OE34	**Electrical and Hybrid Vehicles**	**3-0-0**	**3**

COURSE OBJECTIVES

- This course is an introduction to the fundamental concepts and techniques in Electrical and Hybrid Vehicles for energy efficient transportation systems

COURSE OUTCOMES:

After completion of the course students will be able to
CO1: analyze hybrid vehicles and their performance
CO2: explain the possible ways of energy storage
CO3: analyze the strategies related to energy storage systems

MODULE 1: Fundamentals

Conventional Vehicles: Basics of vehicle performance, vehicle power source characterization, transmission characteristics, mathematical models to describe vehicle performance;
Hybrid Vehicles: History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies. Hybrid Electric Drive-trains: Basic concept of hybrid traction, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies, fuel efficiency analysis

MODULE 2: Electric Trains

Electric Drive-trains: Basic concept of electric traction, introduction to various electric drive train topologies, power flow control in electric drive-train topologies, fuel efficiency analysis. Electric Propulsion unit: Introduction to electric components used in hybrid and electric vehicles, Configuration and control of DC Motor drives, Configuration and control of Induction Motor drives, configuration and control of Permanent Magnet Motor drives, Configuration and control of Switch Reluctance Motor drives, drive system efficiency

MODULE 3: Energy Storage

Energy Storage: Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles, Battery based energy storage and its analysis, Fuel Cell based energy storage and its analysis, Super Capacitor based energy storage and its analysis, Flywheel based energy storage and its analysis, Hybridization of different energy storage devices; Sizing the drive system: Matching the electric machine and the internal combustion engine (ICE); Sizing the propulsion motor, sizing the power electronics, selecting the energy storage technology, Communications, supporting subsystems

MODULE 4: Energy Management Strategies

Energy Management Strategies: Introduction to energy management strategies used in hybrid and electric vehicles, classification of different energy management strategies, comparison of different energy management strategies, implementation issues of energy management strategies. Case Studies: Design of a Hybrid Electric Vehicle (HEV), Design of a Battery Electric Vehicle (BEV)

Textbooks/ Reference Books:

1. C. Mi, M. A. Masrur and D. W. Gao: Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives, John Wiley & Sons, 2011

2. S. Onori, L. Serrao and G. Rizzoni: Hybrid Electric Vehicles: Energy Management Strategies, Springer, 2015
3. M. Ehsani, Y. Gao, S. E. Gay and A. Emadi: Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design, CRC Press, 2004
4. T. Denton: Electric and Hybrid Vehicles, Routledge, 2016

Course Code	Course Title	Hours per week L-T-P	Credit C
HS181704	**Principles of Management**	**3-0-0**	**3**

MODULE1: Introduction
Definition and meaning of management, Characteristics of management, importance of management, functions of management-planning, organising, directing, staffing, coordination and controlling etc., principles of management, Difference between administration and management

MODULE2: Financial Management
Definition and management of financial planning, importance and characteristics of sound financial plan, concepts of capital- fixed capital and working capital, source of finance, fund flow statement.

MODULE3: Marginal costing
Definition and meaning of marginal costing, advantages, marginal cost equation, contribution, profit-volume ratio, break even analysis, margin of safety.

MODULE4: Cost Accounting
Cost Accounting- Concept and benefit, elements of cost, preparation of cost sheet with adjustment of raw materials, work-in-progress and finished goods.

MODULE5: Capitalisation
Definition and meaning of capitalisation, over and under capitalisation.

MODULE6: Motivation
Introductory observation, definition of motivation, motivational technique, features of sound motivational system.

MODULE7: Leadership
Concept of leadership, principles of leadership, functions of leadership, qualities of leadership, different styles of leadership**.**

Textbooks/Reference Books:

1. Principle of Business Management: RK Sharma, Shashi K.Gupta
2. Business Organisation and Management: SS Sarkar, RK Sharma, Shashi K.Gupta
3. Industrial Organisation and Management: SK Basu, KC Sahu, B Rajviv
4. Principles of Management by Dr. A. K. Bora: Chandra Prakash, Guwahati.
5. Management Accounting: RK Sharma, Shashi K Gupta
6. Cost Accounting: SP Jain, K I Narang
7. Cost Accounting, RSN Pillai, V Bhagawati
8. Principles of Management: RN Gupta
9. Principles of Management: RSN Pillai, S. Kala
10. Principles of Management: Dipak Kumar Bhattacharj

Course Code	Course Title	Hours per week L-T-P	Credit C
EE181722	**Project-1**	**0-0-6**	**3**
GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME			

Course Code	Course Title	Hours per week L-T-P	Credit C
SI181721	**Internship-III (SAI - Industry)**	**0-0-0**	**2**
GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME			

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